

CRED-RAPTOR: A Hierarchical RAG-based LLM Pipeline for Corporate Credit Risk Assessment

Anonymous ACL submission

Abstract

Corporate credit risk assessment is a critical process for financial institutions, in accessing the corporate borrowers at a bank. The process is not just a bank’s internal process, it affects the future risk and profit associated with the credit granted. Despite its importance, the current workflow is almost manual. Financial analyst must meticulously review massive, complex corporate documents to evaluate a borrower’s health. This traditional approach is labor-intensive and is prone to human error. Furthermore, standard methods struggle to unlock the full semantic potential of a large document and fail to evaluate new firms with no institutional history.

To address these challenges, we present a novel, automated pipeline that fuses document intelligence with machine learning to streamline credit scoring. Our framework utilizes a hierarchical Retrieval-Augmented Generation (RAG) architecture to effectively parse and comprehend long-form financial documents. Using this, we integrate these qualitative textual features with the traditional quantitative financial metrics. This unified feature space is subsequently processed by a downstream machine learning model to predict the credit score. Crucially, our pipeline addresses the strict demand for auditability by making all the feature sources traceable back to the raw text. We also resolve the "cold-start" problem for new businesses by leveraging data from its industry peers to fill the information gaps. CRED-RAPTOR closely mimics the comprehensive evaluation paradigm of expert financial analysts, serving as an auditable decision-support tool to the financial analysts which can significantly reduce manual labor and human risk.

1 Introduction

Auditability and transparency are the fundamental principles banking institutions strictly rely on. Corporate credit risk evaluation also demands the

same. Traditionally, credit scoring systems and frameworks heavily rely on quantitative metrics such as leverage ratios, profitability measures, and liquidity indicators to predict creditworthiness and performance of a corporate completely missing out on qualitative information. However, in practice, incorporating qualitative information in credit risk assessment, traditionally means manual review of lengthy documents by risk analysts, which is prone to human error and leaves little standardized record of which piece of text drove the given judgement.

General credit scoring models are based on quantitative features of any corporate with expert qualitative analysis. Many Studies have proved the importance of using qualitative features in credit decision making. [Soares et al. \(2011\)](#) surveyed loan officers across six major Portuguese banks and found that qualitative factors such as management experience and reliability were the most valued considerations in credit decision making, and their study also found that banks placing more emphasis on qualitative factors tended to have fewer overdue loans.

Even when banks do formalize this qualitative review through disclosure, its consistency and depth vary widely. A qualitative content analysis of Canadian and Italian bank disclosures found that although the two countries’ overall disclosure quality scores were not significantly different, yet Italian banks tended to provide more granular, comprehensive disclosure while Canadian banks offered a more holistic view of credit risk, with the completeness of a bank’s pillar 3 report emerging as a key driver of the disclosure quality score ([Lajili et al., 2022](#)). This variation underscores a broader problem: manual qualitative credit assessment lacks a consistent, auditable methodology for tracing a score back to its source text, which is precisely the gap that automated, source-grounded extraction methods aim to close.

In recent years, the integration of artificial in-

telligence and machine learning into credit risk assessment is showing good results, with studies reporting improvements of up to 30% in prediction accuracy and 40% in default rate reduction (Dil and Ramchand, 2025). Early approaches for credit rating forecasting using machine learning compared neural networks, support vector machines, and decision trees directly against one another and showed that ensemble and tree-based methods worked well, balancing accuracy and stability, for structured financial ratio data (Golbayani et al., 2020), a result that later studies supported, as gradient boosting methods such as XGBoost became a common baseline approach. Following this trend, LLMs have begun reshaping how qualitative financial information is processed. One recent framework combines LLM-based reasoning with gradient-boosted models like CatBoost, LightGBM, and XGBoost to improve both accuracy and interpretability in credit scoring, and it was tested using ratings from several major agencies (Majumdar et al., 2025).

A systematic review of sixty papers published between 2020 and 2025 has documented the quickly expanding space of LLM-based credit risk estimation, especially looking at explainability approaches like chain-of-thought prompting and natural language justification (Golec and AlabdulJalil, 2025). Another work has applied LLMs directly to financial credit tasks by fine-tuning them with instructions across a range of datasets (Lei et al., 2025), while other comparative studies have tested prompt-based LLM classifiers against traditional models like XGBoost, support vector machines, random forests, and multi-layer perceptron’s on standard credit risk datasets (Chen, 2025). Closer to our own data source, one framework used LLMs to rate companies directly from their 10-K filings, across dimensions like sustainability, innovation, and workforce management, showing that these filings alone can support scalable company evaluation without manual review (Daimi and Iqbal, 2024).

Collectively, these studies show that LLMs are not just text generation tools but are efficient in extracting structured, decision relevant signals from financial texts. Even with these advances, the “black box” nature of LLMs poses interpretability problems that go against banks’ regulatory need for auditable and transparent credit decisions. Hence, the black-box part is only used to pull out features from long documents, while interpretable methods take care of the final scoring.

Large language models also impose several problems including hallucinations etc. Retrieval Augmented Generation (RAG) helps here by grounding extracted features in verifiable source documents, supporting auditability (Rao et al., 2025). This grounding also improves accuracy since LLMs answer questions over bank financial reports, by anchoring responses in retrieved evidence rather than memory alone (Iaroshev et al., 2024).

Though RAG allows to ground the responses but it also extracts the retrieved content upto a limit which may not work efficiently with documents where information is scattered along the document. Advanced RAG architectures like RAPTOR solves this problem by building a tree of document summaries by clustering them recursively, allowing retrieval at different levels of detail, and this performs better on knowledge-heavy tasks than simple chunk-based retrieval (Sarathi et al., 2024). One adaptation of RAPTOR for financial documents replaces its text-similarity clustering with clustering by company, sector, and year, since financial reports can be read very similarly in language while describing very different financial realities. This metadata-based clustering produced noticeably better answer quality on multi-document financial questions than either plain retrieval or the original RAPTOR (Chawla and Naliath, 2024). Hybrid systems that integrate RAG based extraction with predictive models further ensure that downstream predictions remain traceable to the retrieved information that supports them (Mittapelly, 2024).

A related challenge is also making sure whatever an LLM extracts from filings maps consistently onto a fixed set of categories, rather than producing inconsistent, one-off labels. One study pipeline tackled this for risk factor extraction from 10-K filings, an LLM first extracts risks with supporting quotes, embeddings then map these to a predefined taxonomy, and a second LLM validates each match to filter out poor fits. The system also let an AI agent auto-refine weak taxonomy categories over time. Tested on S&P 500 filings, it showed that the extracted risk categories reflected real industry patterns, even though the system never saw industry codes (Dolphin et al., 2026). This same challenge, mapping messy LLM output onto a consistent, auditable set of categories, applies directly to our own task of mapping qualitative credit features onto a fixed scoring rubric.

Separately, a long line of accounting research shows that the narrative parts of 10-K filings, es-

pecially the Management Discussion and Analysis and risk factor sections, carry useful information about a firm’s financial distress that goes beyond what accounting ratios alone can show. Studies that combine accounting, trading, and text data have found that NLP techniques can turn this unstructured text into useful predictive features, with deep learning methods improving bankruptcy prediction compared to accounting-only baselines (Mai et al., 2019).

A study also has found that textual disclosures can distinguish bankrupt from non-bankrupt firms, and that the tone of MD&A sections carries real explanatory power for corporate failure (Arno et al., 2022). A follow-up study combining structured financials with textual disclosures also tested zero-shot bankruptcy prediction using GPT-based models, finding that while LLMs summarized the text well, zero-shot prediction alone remained weak without a dedicated predictive model (Arno et al., 2023), supporting our own choice to separate LLM-based feature extraction from prediction rather than asking the LLM to score companies directly. Similarly, aspects like the length, content, and tone of risk disclosures are found to have been linked to a firm’s default risk the following year (Jin et al., 2024). Together, these findings support using filings like 10-Ks, which closely resemble the credit memos analysts already review, as a rich, under-used source of qualitative signal for automated credit assessment.

This study builds on these threads, hierarchical RAG-based extraction, taxonomy-aligned mapping of LLM output, and machine learning-based credit scoring, by combining a hierarchical RAG framework also known as RAPTOR with an LLM for qualitative feature extraction, and XGBoost for credit score prediction. It also introduces a clustering-based peer approach for cold-start companies with no prior filing history, a problem most existing LLM-based credit scoring studies overlook by assuming every borrower already has an established filing or rating history. has an established filing or rating history.

2 Related Work

The most similar work to this study is the LLM-driven framework that combines LLM-based qualitative reasoning with gradient-boosted models like CatBoost, LightGBM, and XGBoost for corporate credit scoring across multiple rating agencies (Ma-

jumdar et al., 2025). It shares our study’s core idea of pairing LLM-extracted signal with tree-based prediction, but treats the LLM’s output as a general reasoning input rather than constraining it to a fixed, auditable set of dimensions, and it also does not address companies with no prior filing history. Our study is also closely related to a pipeline that maps free-form LLM-extracted risk factors from 10-K filings onto a fixed taxonomy using embeddings and a second LLM as a judge (Dolphin et al., 2026). This addresses a similar problem—keeping LLM output structured and auditable—but for open-ended risk categorization rather than credit scoring itself, and via post-hoc mapping rather than constraining the extraction prompt directly.

This study differs from both in three key ways: it constrains the LLM at extraction time to score five predefined credit risk dimensions using hierarchical RAPTOR retrieval (Sarathi et al., 2024), rather than mapping open output after the fact or leaving it unconstrained; it fuses these qualitative scores with programmatic quantitative KPIs and an explicit source indicator, rather than relying on qualitative signal alone; and it explicitly handles cold-start companies through KNN-based peer clustering, a case neither prior work considers.

3 Methodology

We propose a hybrid pipeline for corporate credit risk scoring including an efficient use of black box model while maintaining auditability and transparency in the pipeline. Credit scoring framework relies on an XGBoost classifier which is fed 2 types of features: (1) Quantitative features, programmatically extracted from the SEC filings’ financial reports from SEC EDGAR, (2) Qualitative features generated via a Recursive Abstractive Processing for Tree-Organized Retrieval (RAPTOR) on SEC filings, while a third feature which is a binary indicator or SOURCE INDICATOR, helps differentiating the cold start companies from companies whose SEC Filings are available. The complete end-to-end architecture is illustrated in Figure 1.

3.1 Data Sources and Target Variable

We formulate credit risk assessment as a multi-class classification problem. The target variable is the standardized corporate credit rating, sourced directly from Egan-Jones Ratings Company XML files. Egan-Jones provides independent, institutional-grade credit ratings which we

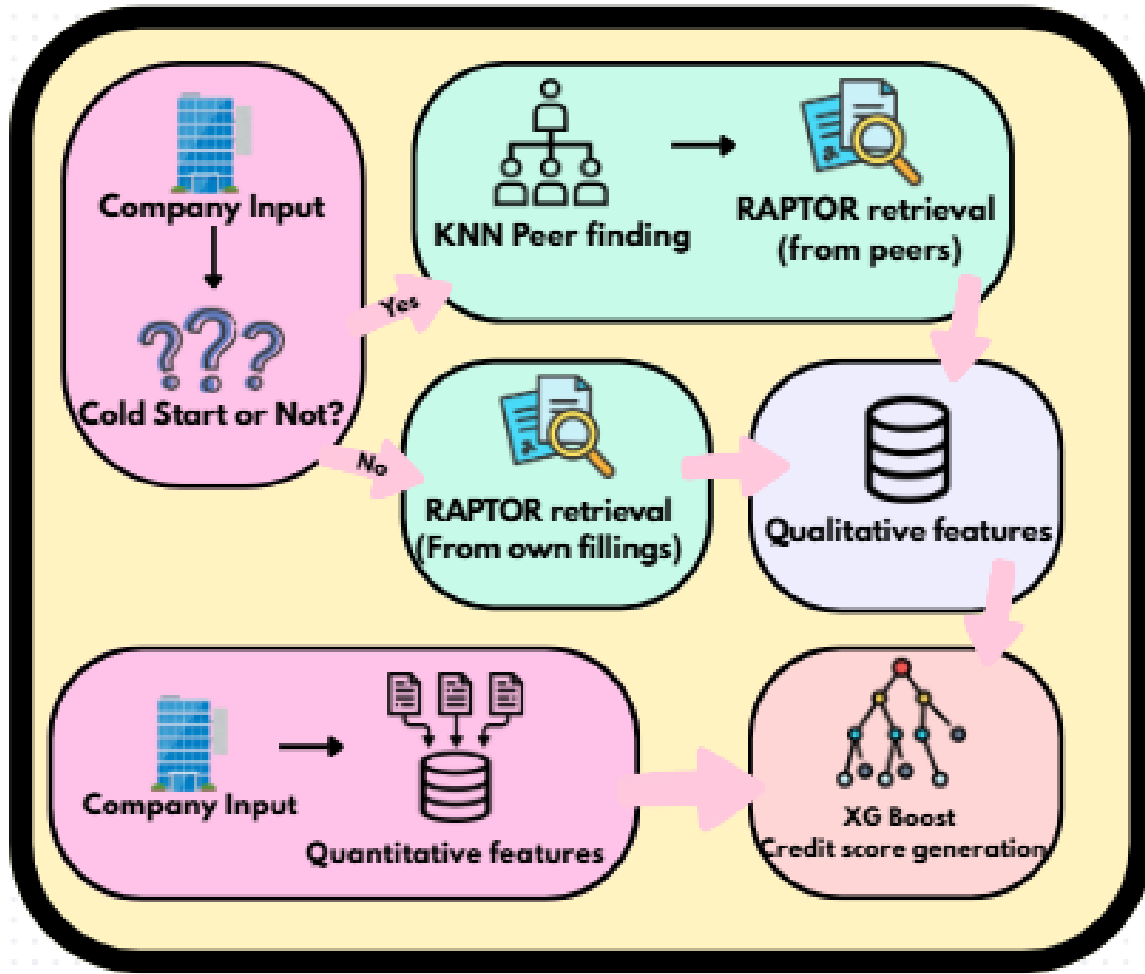


Figure 1: The proposed tri-modal credit scoring architecture. Quantitative KPIs are merged with qualitative sentiment features extracted via hierarchical RAPTOR tree-summarization to form the final feature vector for the XGBoost classifier.

programmatically parse and map onto a granular 21-point ordinal numeric scale, ranging from AAA represented numerically as 20 down to default grades D as 0.

We curated the corresponding annual financial reports (Form 10-K) for these rated entities from the SEC EDGAR database, yielding a final valid dataset of 2,349 paired financial reports and credit ratings, representing 245 unique corporate entities. Form 10-K filings were specifically selected because they provide both the standardized financial tables required for our quantitative baselines and the forward-looking management narratives.

3.2 Quantitative Feature Calculation

To empirically assess the Credit Risk, we leverage five robust financial ratios adapted from the classical Altman Z-Score framework for bankruptcy prediction (Altman, 1968) and the Ohlson O-Score model for default probability (Ohlson, 1980).

These ratios are programmatically extracted and calculated from the standardized financial tables within the SEC 10-K filings. The eight KPIs and their financial rationales are as follows:

- Debt-to-Equity:** Measures financial leverage and the proportion of shareholders' equity and debt used to finance a company's assets.
- Retained Earnings / Total Assets:** Serves as a proxy for the company's age and cumulative profitability.
- Current Ratio:** Evaluates a company's ability to pay short-term obligations (due within one year) by comparing current assets to current liabilities.
- Quick Ratio:** A measure of liquidity, evaluating immediate solvency by excluding inventory and other less liquid current assets.

5. **Working Capital / Total Assets:** Measures the short-term liquidity of the firm relative to its total capitalization.
6. **ROCE (Return on Capital Employed):** Indicates how efficiently a company is using its capital base to generate operating profits.
7. **Net Profit Margin:** Measures how much net income is generated as a percentage of revenue, indicating overall profitability.
8. **Total Liabilities / Total Assets:** Evaluates the overall long-term solvency of the company.

3.3 Qualitative Feature Extraction

As depicted in Figure 2, the qualitative extraction process is bifurcated on the basis of the availability of historical filing data. For established companies, their own SEC 10-K filings are ingested directly. However, if a company is identified as a "cold-start" meaning it lacks sufficient public filing history, we use a K-Nearest Neighbors (KNN) algorithm to identify the K most similar peer companies. The SEC filings of these peer companies are subsequently aggregated to serve as a proxy for the cold-start entity’s qualitative baseline.

These filings are processed using RAPTOR (Recursive Abstractive Processing for Tree-Organized Retrieval) (Sarathi et al., 2024). It is chosen over flat chunking approaches because it captures both fine-grained detail and broad thematic context in a single retrieval structure, an important property given that credit-relevant qualitative signals can appear either as a specific line in a risk disclosure or as a pattern spread across an entire filing. The extraction pipeline functions as follows:

1. **Segmentation and Embedding:** The SEC filing is segmented into contiguous text chunks of 512 tokens. These chunks are embedded into a dense vector space using yixuantt/Fin-E5, a state-of-the-art embedding model specifically tuned to capture financial semantic nuances better than generalized models (Tang and Yang, 2025).
2. **Recursive Clustering:** The chunk embeddings are softly clustered using Gaussian Mixture Models (GMM), allowing semantically related texts to group together. A single chunk may belong to multiple clusters, preserving multifaceted contexts.

3. **Hierarchical Summarization:** Each distinct cluster is passed to a localized Llama-3.1-8B-Instruct model to generate a comprehensive parent summary. These new summary nodes are then re-embedded and re-clustered recursively, building a bottom-up tree until convergence into a single root node.

4. **Retrieval and Scoring:** At inference time, both the raw leaf nodes and the abstractive summary nodes are retrieved via cosine similarity.

The retrieved context is then passed to the same Llama-3.1-8B-Instruct model, prompted to perform sentiment and risk extraction across five credit dimensions on a standardized continuous scale:

1. **Revenue Sentiment:** Drivers of sales growth, contraction, and high-level market trends.
2. **Operating Profit Sentiment:** Factors impacting operational efficiency and core business profitability.
3. **Net/Gross Margins Sentiment:** The effects of pricing power, inflation, and cost of goods sold (COGS).
4. **Net Profit Sentiment:** Management’s narrative regarding bottom-line earnings, taxes, and exceptional expenses.
5. **Free Cash Flow Sentiment:** Commentary surrounding cash generation, capital expenditures, debt repayment, and liquidity constraints.

Rather than relying on the LLM to directly assign a numerical sentiment label which can lead to malformed outputs or hallucinated categories. The extracted text for each dimension is subsequently passed to ProsusAI/finbert, which is heavily pre-trained on financial corpora (Araci, 2019), computes a continuous sentiment score for each dimension. This continuous score S_{dim} is calculated as the probability of a positive sentiment minus the probability of a negative sentiment:

$$S_{dim} = P_{pos} - P_{neg} \quad (1)$$

This calculation yields a standardized continuous scale bounded between -1.0 and 1.0. When the LLM fails to extract grounded text for a specific dimension, a neutral sentiment score of 0.0 is assigned as a fallback rather than discarding the observation.

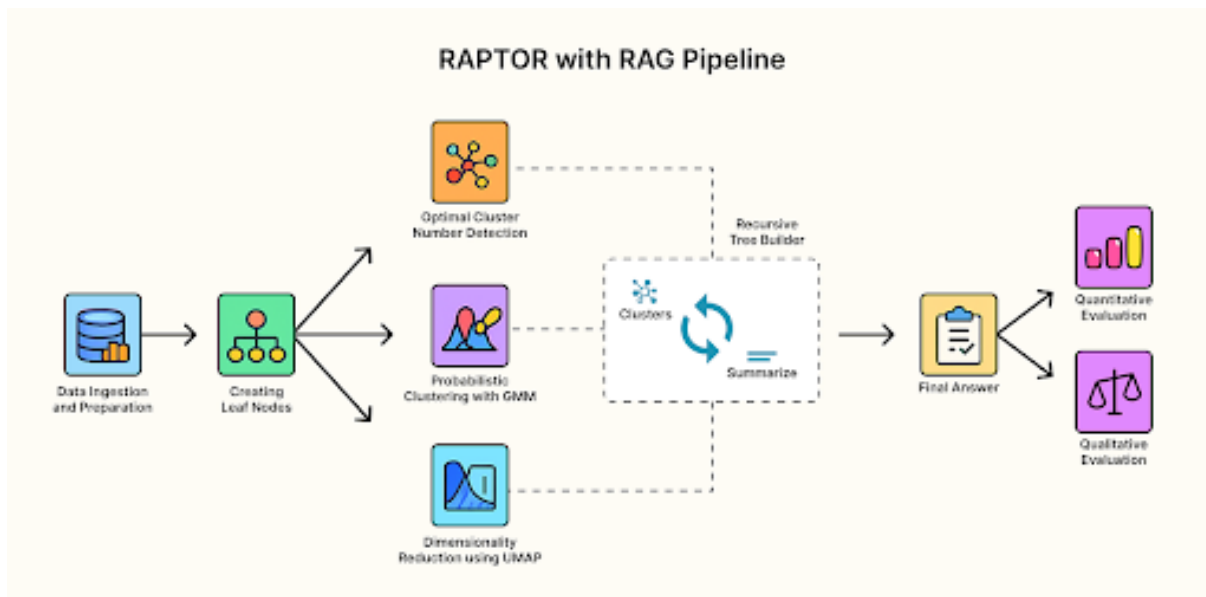


Figure 2: The RAPTOR hierarchical retrieval mechanism. Financial 10-K filings are softly clustered via GMM and recursively summarized by an LLM to build a structural semantic tree, preserving both isolated facts and overarching macroeconomic risks.

3.4 Handling Cold-Start Companies

For companies with no prior institutional filing history, qualitative assessment fundamentally fails. To address this gap, we employ a K-Nearest Neighbors (KNN) algorithm, configured using $K = 5$ and Euclidean distance mapping on industry features, the algorithm identifies peer companies so their filings can be aggregated and used in place of the cold-start entity’s missing reports.

To ensure complete transparency, we introduce a binary source indicator. This feature records whether a company’s qualitative and quantitative features were derived from its own SEC filings or from a peer group’s proxy filings. Crucially, this indicator is not fed into the downstream predictive model as a learned feature; rather, it provides a strict, immutable audit trail for risk analysts, ensuring it is always clear exactly which companies were scored on borrowed information versus verified historical filings.

3.5 Traceability and Auditability

A critical advantage of our hierarchical architecture is strict auditability, mitigating the "black-box" nature of LLMs. By decoupling extraction from prediction, our pipeline provides a deterministic, backward-chaining traceback mechanism. If an analyst observes an XGBoost downgrade driven by a negative Revenue_Sentiment, they can query the intermediate pipeline logs to retrieve the exact

qualitative summary generated by Llama-3.1 that yielded that score. From there, the RAPTOR retrieval logs trace that summary directly back to the specific text chunks extracted from the raw SEC 10-K filing. This structured separation guarantees that human analysts can systematically verify the exact linguistic and financial rationale behind any algorithmic credit decision.

4 Experimental Setup

To ensure robust evaluation across the highly imbalanced 21-notch rating scale, the dataset of 2,349 filings was partitioned into training (80%) and testing (20%) sets. We utilized stratified sampling based on the ground-truth credit rating labels to guarantee that minority classes were proportionally represented in both the training and validation set.

All qualitative feature extraction was conducted using a localized pipeline to ensure data privacy a critical requirement for financial institutions. Semantic embeddings were generated using the yixuantt/Fin-E5 model. For the RAPTOR pipeline, filings were segmented into base chunks of 512 tokens with a 50-token overlap. The hierarchical clustering and abstractive summarization were processed locally using meta-llama/Llama-3.1-8B-Instruct deployed via the llama.cpp inference engine. The LLM extraction prompt was configured with a highly deterministic base temperature of 0.1 and a generation

Model Strategy	Macro Accuracy	MAE	Within-1-Bucket	Weighted F1
Baseline (Quant Only)	58.51%	0.618	83.82%	0.5966
Naive RAG (Quant+Flat Text)	54.17%	0.667	79.17%	0.5378
RAPTOR Hybrid (Quant+Tree Text)	63.07%	0.518	88.38%	0.6371

Table 1: Model Performance on 6-Bucket Macro Scale (Temporal Split, 80/20). RAPTOR significantly outperforms the quantitative baseline across all metrics.

limit of 2,048 tokens. Final sentiment scoring was performed using the ProsusAI/finbert sequence classification model, running in inference batches of 128 with a truncated maximum sequence length of 512 tokens.

For the downstream predictive model, we utilized the XGBoost framework. Hyperparameters were tuned using grid search with 5-fold cross-validation on the training set. The optimal configuration utilized a learning rate (η) of 0.05, a maximum tree depth of 6, and 300 estimators, with the multi-class softmax objective function.

Because credit ratings are inherently ordinal and highly imbalanced, relying on raw accuracy can be misleading. Therefore, we evaluate all models using both standard **Accuracy** and the **Weighted F1-Score**.

5 Results

To evaluate the predictive power of our proposed architecture, we benchmarked our hierarchical retrieval approach against two baseline models. Each model was evaluated using standard multi-class classification metrics to ensure robust measurement across the 21 credit rating scale. The three evaluated pipelines are as follows:

- **Quantitative Baseline:** A standard XGBoost classifier trained strictly on the eight programmatic financial KPIs, representing the traditional, numbers-only approach to credit scoring.
- **Naive RAG Hybrid:** An XGBoost classifier trained on both quantitative KPIs and qualitative sentiment scores extracted via a traditional, flat-chunking RAG pipeline.
- **RAPTOR RAG Hybrid (Proposed):** An XGBoost classifier trained on the quantitative KPIs and qualitative sentiment scores extracted via our hierarchical RAPTOR pipeline, representing full contextual grounding.

5.1 Comparative Pipeline Performance

As shown in Table 1, the traditional Naive RAG approach actually degraded predictive performance compared to the Quantitative Baseline, dropping Accuracy from 58.51% to 54.17%. This empirical degradation highlights a critical flaw in applying standard flat-chunking retrieval to complex financial documents.

In contrast, the proposed RAPTOR-based extraction significantly outperformed both baselines, achieving a peak Accuracy of 63.07% and a Weighted F1-Score of 0.6371. This validates our hypothesis that hierarchical tree summarization successfully captures broad, filing-wide thematic risk factors that flat chunking misses.

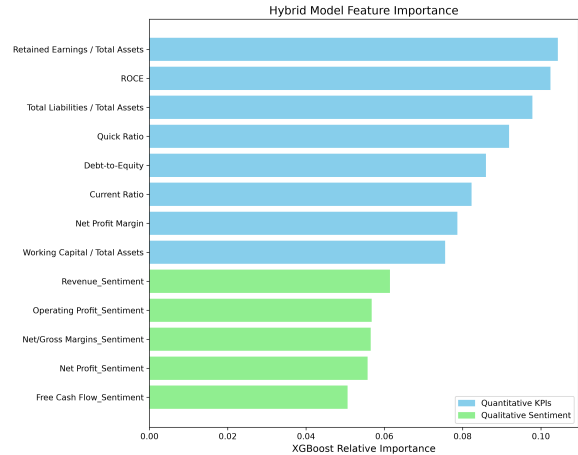


Figure 3: XGBoost Feature Importance for the RAPTOR Hybrid Pipeline. Note the significant contribution of the LLM-derived qualitative sentiment dimensions alongside traditional financial KPIs.

5.2 Ablation Studies

To ensure the robustness of our architecture, we conducted targeted ablation studies on the two most critical components of the pipeline: the hierarchical text extraction and the cold-start peer proxying.

5.2.1 RAPTOR vs. Naive RAG Extraction

As demonstrated in the primary results in Table 1, Naive RAG actively degrades the model’s predic-

Abstraction	Pipeline	Precision	Recall	Relevance
Conceptual	Naive RAG	4.00	3.20	2.50
	RAPTOR RAG	4.40	3.70	2.80
Strategic	Naive RAG	3.70	2.50	2.70
	RAPTOR RAG	3.40	2.70	2.60

Table 2: Retrieval evaluation focusing on complex cognitive abstraction levels. RAPTOR demonstrates superiority in synthesizing the broad Conceptual contexts required for credit scoring.

tive power below the quantitative baseline. Financial narratives in 10-K filings are highly contextual. In contrast, the RAPTOR pipeline forces the LLM to construct a bottom-up semantic tree, effectively summarizing the filing’s risk profile before generating a sentiment score. The 8.9% absolute accuracy jump from Naive RAG to RAPTOR RAG empirically validates this architectural choice.

To further isolate and validate this retrieval architecture, we conducted an evaluation of the generative retrieval pipelines using a 30-question diagnostic dataset split evenly across Conceptual and Strategic abstraction levels scored by an LLM-as-a-Judge on a 1-5 scale.

As shown in Table 2, the traditional flat-chunking pipeline struggles when tasked with Conceptual risk factors. While Naive RAG maintains a slight edge in Strategic precision, RAPTOR consistently outperforms it across Conceptual metrics. Because credit risk analysis inherently relies on synthesizing broad, filing-wide themes and the Naive RAG starves the downstream XGBoost classifier of vital context. Conversely, the RAPTOR hierarchical tree structure successfully preserves these overarching narratives, dominating the Naive baseline in Conceptual precision, recall, and relevance.

5.2.2 Cold-Start Evaluation

To address the cold-start problem for companies lacking 10-K filings, we explored a K-Nearest Neighbors (KNN) peer-proxy mechanism. This approach maps a cold-start company to its 5 closest quantitative peers and borrows their averaged RAPTOR sentiment embeddings. However, our empirical evaluation in a strictly Zero-Shot setting yielded only 41.82% which is a significant drop from our main model’s 63.07%.

6 Conclusion

In this paper, we presented CRED-RAPTOR, a novel pipeline that automates corporate credit risk assessment by combining traditional financial numbers with extracted qualitative sentiment. Testing our system on 2,349 annual filings showed that standard retrieval methods predictive performance by stripping vital context. In contrast, our hierarchical RAPTOR architecture successfully captures broad, filing-wide risk themes. As a result, CRED-RAPTOR achieved a peak macro-bucket accuracy of 63.07% and correctly placed companies within one rating bucket 88.38% of the time, easily outperforming traditional quantitative baselines.

Furthermore, our cold-start experiments proved that a company’s qualitative risk is highly unique, you cannot simply borrow text from a quantitative peer to predict a score. This reinforces why analyzing a company’s actual disclosures is absolutely necessary. Most importantly, our pipeline achieves this strong performance while maintaining a deterministic, backward-chaining audit trail from the final credit score all the way back to the raw SEC text. This solves a major "black-box" challenge, bridging the gap between advanced Large Language Models and the strict interpretability requirements of modern financial institutions.

7 Limitations

While our pipeline demonstrates strong performance, there are limitations that exist. First, our dataset of 2,349 annual filings is not substantial and small compared to general-domain NLP datasets, potentially limiting the generalizability of the XGBoost models across unseen macroeconomic regimes. Second, our pipeline relies on the Llama-3.1-8B-Instruct model for qualitative extraction. Although using an open-weights model ensures reproducibility and data privacy, it remains susceptible to hallucination or inconsistent extrac-

tion than a fine-tuned foundational model further stabilize this qualitative extraction phase.

8 Ethics and Broader Impact

The automation of credit risk scoring carries significant ethical and regulatory implications. Algorithmic systems must comply with fair lending laws and avoid disparate impact. While corporate credit assessment does not directly evaluate individual demographics, systemic biases in historical rating agency data could inadvertently disadvantage smaller enterprises if encoded into the LLM representations. Furthermore, our findings regarding the idiosyncratic nature of qualitative risk highlight the ethical dangers of proxying financial sentiment. Attempting to algorithmically infer a cold-start company’s risk profile purely from its quantitative peers can perpetuate industry-wide biases and unjustly penalize disruptive innovation. Any deployment by financial institutions must ensure humans remain in the loop, utilizing the model’s auditability features to verify the rationale behind rating downgrades.

References

- E. I. Altman. 1968. Financial ratios, discriminant analysis and the prediction of corporate bankruptcy. *The Journal of Finance*.
- Dogu Araci. 2019. Finbert: Financial sentiment analysis with pre-trained language models. *arXiv preprint arXiv:1908.10063*.
- H. Arno, K. Mulier, J. Baeck, and T. Demeester. 2022. Next year bankruptcy prediction from textual data: Benchmark and baselines. In *Proc. 4th Workshop on Financial Technology and Natural Language Processing (FinNLP), IJCAI ECAI, Vienna, Austria*.
- H. Arno, K. Mulier, J. Baeck, and T. Demeester. 2023. From numbers to words: Multi modal bankruptcy prediction using the ecl dataset. In *Proc. Sixth Workshop on Financial Technology and Natural Language Processing (FinNLP)*, pages 11–21, Bali, Indonesia.
- K. Chawla and A. Naliath. 2024. Finrag: A retrieval based financial analyst. Technical report, Stanford University CS224N Course Project Report.
- Q. Chen. 2025. Explore the use of prompt based llm for credit risk classification. *Journal of Financial Risk Management (SCIRP)*.
- S. A. Daimi and A. Iqbal. 2024. A scalable data driven framework for systematic analysis of sec 10 k filings using large language models. *arXiv preprint arXiv:2409.17581*.
- P. Dil and A. Ramchand. 2025. Ai and machine learning in credit risk assessment. *SSRN Working Paper*.
- R. Dolphin, J. Dursun, J. Blankenship, K. Adams, and Q. Pike. 2026. Taxonomy aligned risk extraction from 10 k filings with autonomous improvement using llms. *arXiv preprint arXiv:2601.15247*.
- P. Golbayani, I. Florescu, and R. Chatterjee. 2020. A comparative study of forecasting corporate credit ratings using neural networks, support vector machines, and decision trees. *arXiv preprint arXiv:2007.06617*.
- M. Golec and M. AlabdulJalil. 2025. Interpretable llms for credit risk: A systematic review and taxonomy. *Expert Systems with Applications*.
- I. Iaroshev, R. Pillai, L. Vaglietti, and T. Hanne. 2024. Evaluating retrieval augmented generation models for financial report question and answering. *Applied Sciences*, 14(20):9318.
- S. C. Jin, H. W. Lee, and Y. A. Xie. 2024. Risk disclosures in annual reports and firms’ default risk. *Journal of Corporate Accounting & Finance*, 35:192–212.
- K. Lajili, S. Mohsni, S. Polizzi, and E. Scannella. 2022. A qualitative analysis of bank credit risk disclosure: Evidence from the canadian and italian banking sectors. *Journal of Corporate Accounting & Finance*, 33:207–230.
- Y. Lei, Z. Wang, C. Liu, and T. Wang. 2025. Zigong 1.0: A large language model for financial credit. *arXiv preprint arXiv:2502.16159*.
- F. Mai, S. Tian, C. Lee, and L. Ma. 2019. Deep learning models for bankruptcy prediction using textual disclosures. *European Journal of Operational Research*, 274(2):743–758.
- C. Majumdar, S. Scandizzo, R. Mahanta, A. Mandal, and S. Bhattacharjee. 2025. A large language model for corporate credit scoring. *arXiv preprint arXiv:2511.02593*.
- A. K. Mittapelly. 2024. Enhancing credit risk analysis in financial crms using retrieval augmented generation and predictive ai. *J. Artif. Intell. Mach. Learn. & Data Sci.*, 2(3):2617–2623.
- J. A. Ohlson. 1980. Financial ratios and the probabilistic prediction of bankruptcy. *Journal of Accounting Research*, 18(1):109–131.
- P. S. Rao, S. Ščepanović, K. Zhou, E. P. Bogucka, and D. Quercia. 2025. Riskrag: A data driven solution for improved ai model risk reporting. In *Proc. 2025 CHI Conference on Human Factors in Computing Systems*, pages 1–26.
- P. Sarthi, S. Abdullah, A. Tuli, S. Khanna, A. Goldie, and C. D. Manning. 2024. Raptor: Recursive abstractive processing for tree organized retrieval. In *The Twelfth International Conference on Learning Representations (ICLR)*.

J. O. Soares, J. Pina, M. Ribeiro, and M. C. Lopes. 2011. Quantitative vs. qualitative criteria for credit risk assessment. *Frontiers in Finance and Economics*, 8(1):69–87.

Yixuan Tang and Yi Yang. 2025. Finmteb: A massive text embedding benchmark for the financial domain. *arXiv preprint*.

A Appendix

A.1 LLM Extraction prompt

To ensure strict reproducibility and minimize multi-task hallucination, we did not ask the model to extract all risk dimensions simultaneously. Instead, we instantiated a dedicated, parameterized prompt for each of the five target dimensions: Revenue, Operating Profit, Net/Gross Margins, Net Profit, and Free Cash Flow.

The Llama-3.1-8B-Instruct model was iteratively queried using the following prompt structure, where [DIMENSION] was replaced by the specific risk category being evaluated:

System: You are an expert corporate credit risk analyst conducting a rigorous financial audit. I will provide you with a comprehensive summary of a company’s SEC 10-K filing. Your task is to perform targeted qualitative extraction for exactly one specific financial dimension: [DIMENSION].

Carefully analyze the text and extract any management commentary, forward-looking guidance, or identified risk factors that directly impact this specific dimension. Your extraction must be entirely grounded in the provided text. Do not summarize other metrics or hallucinate external macroeconomic factors.

If the provided summary contains no relevant information regarding [DIMENSION], you must output exactly the word "None".

User: [Insert RAPTOR Summary Text]